# Mohsena Ashraf

**PhD Student**, Computer Science
University of Colorado Boulder, USA

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## EDUCATION

**PhD in Computer Science** *[Fall 2022 – Present]*

**University of Colorado Boulder**, USA

**Graduate Assistant** *[Fall 2022-Present]*

**MSc in Computer Science** *[Fall 2022 – Fall 2024]*

**University of Colorado Boulder**, USA

**BSc in Computer Science & Engineering** *[2015 – 2019]*

**Ahsanullah University of Science and Technology** Bangladesh

CGPA: **3.936 out of 4.00**

Merit position: **1st out of 152 students**

## RESEARCH INTERESTS

- Agentic AI
- Computer Vision
- Human Computer Interaction
- Machine Learning

## RESEARCH EXPERIENCE

- **Doctoral Researcher - CU Boulder** ***[2022 - Present]***:

  **Agentic AI & Human-Centered AI** *[2024 - Present]*
  Conducting research on making complex agentic AI concepts, architectures, and workflows more understandable and accessible through visual, interactive, and familiar representations. Developing visual representations of agent architectures and workflows that help users understand the underlying components and logic of agentic systems rather than treating them as black boxes. Additionally, developing Excel-based and interactive models to explain AI token usage and computational cost, including how context growth, iterative loops, tool use, and

multi-agent interactions contribute to cost. Investigating the learning needs and challenges of professionals and engineers working with agentic AI to inform the design of more accessible approaches for understanding and building AI agents.

**AI & Parenting Studies** *[2024]*
Collaborated with CU Denver on a study where parents and children co-create stories with a ChatGPT-powered chatbot in multiple settings, exploring parent–child–AI interaction patterns. Using functional near-infrared spectroscopy (fNIRS) to analyze children's brain activity during these creative tasks (Published in *IDC 2023*).

**Research Assistant | fNIRS & Empathy Studies** *[2023 - 2024]*
Worked on fNIRS-based analysis to examine adults' empathy toward children in AI and cybersecurity scenarios. Developed experimental vignettes, collected participant data, and conducted neural data analysis (Published in *CHI EA 2024*).

- **Research team member** *[2020 – 2022]*:

  A project initiated by *CRID-DAM Robotic Innovation Lab (CDRiL),USA* in collaboration with the *Dhaka North City Corporation* that aims to build a robotic assistant for cleaning the drains and surface areas of Bangladesh.

## BSc THESIS

### Brain Tumor Detection using Convoutional Neural Network *[June 2019]*

Developed a system for automated brain tumor detection using MRI and CT images through image processing and machine learning techniques. Utilized the BRATS dataset, and applied methods such as K-means, Fuzzy C-means, six traditional classifiers, and a CNN model to compare traditional and deep learning approaches for tumor identification.

## TEACHING EXPERIENCE

### Graduate Teaching Assistant

University of Colorado Boulder.

**Courses:**

Data Mining (CSCI 4502/5502) — *Fall'24, Fall'25, Fall'26*
Computer Vision (CSCI 4722/5722) — *Spring'23, Fall'23, Spring'24, Spring'25*
Fundamentals of HCI (CSCI 3002) — *Spring'26*

### Graduate Research Assistant

University of Colorado Boulder. — *Fall'25*

### Lecturer, Computer Science and Engineering

Ahsanullah University of Science and Technology. *[August 2019 – July 2022]*

**Courses Taught:**

Digital Image Processing — *Fall'21, Fall'20, Spring'20, Fall'19*

Digital System Design — *Fall'21, Spring'21, Fall'20, Spring'20*
Object Oriented Programming — *Fall'20, Fall'19*
Structured Programming Language (C) — *Fall'21, Fall'19, Spring'19*
Software Development - IV (PHP) — *Spring'20*
Software Development - II (Java) — *Spring'19*

# PUBLICATIONS

[**Google scholar profile:** *here*]

- Sami Rashid, Md Fahim Shahriar, Sumaiya Karim Katha, Ezharuddin Jubaer, Akm Moshiur Rahman Mazumder, Ashraful Islam, **Mohsena Ashraf**, Asif Mahmud, M. Ashraful Amin, *AI Agents in Healthcare: A Survey of Applications, Resources, and Challenges*, IEEE Student Conference on Research and Development (SCOReD) 2025, October 2025 **(Accepted)**.
- Jenna H. Chin, Seungwook Lee, **Mohsena Ashraf**, Matt Zago, Yun Xie, Elizabeth A. Wolfgram, Tom Yeh, and Pilyoung Kim, *Young Children's Creative Storytelling with ChatGPT vs. Parent: Comparing Interactive Styles*, CHI EA '24: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, May 2024.
- **Mohsena Ashraf**, *A systematic review on the potential of AI and ChatGPT for parental support and child well-being*, PhD Comprehensive Exam Report, University of Colorado Boulder, May 2024.
- **Mohsena Ashraf**, Genevieve Patterson, Zachary Kilhoffer, Xu Han, Nolan Brady, Anna Rahn, Nikhitha Atluri, Violet Oliver, Yun Huang, Yang Wang, Pilyoung Kim, and Tom Yeh, *Using fNIRS To Understand Adults' Empathy for Children in AI and Cybersecurity Scenarios*, IDC '23: Proceedings of the 22nd Annual ACM Interaction Design and Children Conference, June 2023, **(Works-in-Progress Paper)**.
- **Mohsena Ashraf**, Farzana Anowar, Jahanggir Hossain Setu, Atiqul Islam Chowdhury, Eshtiak Ahmed, Ashraful Islam, and Abdullah Al-Mamun, *A Survey on Dimensionality Reduction Techniques for Time-series Data*, IEEE Access, April 2023. [impact factor 3.476]
- **Mohsena Ashraf**, *Designing vignettes for fNIRs studies for Analyzing Adults' Empathy for Children in Technology: An Ethical Perspective*, Submitted at The ACM Interaction Design and Children Conference 2023- *'Participatory Approaches to the Ethics of Emerging Technologies for Children'* Category **(Workshop Position Paper)**.
- Mohammad Imrul Jubair, Arafat Ibne Yousuf, Tashfiq Ahmed, Hasanath Jamy, Foisal Reza, and **Mohsena Ashraf**, *Diy Graphics Tab: A Cost-Effective Alternative to Graphics Tablet for Educators*, Accepted at The ACM Symposium on User Interface Software and Technology **(UIST 2022 - Poster Paper)**. [core rank A*]
- Mohammad Imrul Jubair, Arafat Ibne Yousuf, Tashfiq Ahmed, Hasanath Jamy, Foisal Reza, and **Mohsena Ashraf**, *Diy Graphics Tab: A Cost-Effective Alternative to Graphics Tablet for Educators*, Accepted at AAAI Conference on Artificial Intelligence **(AAAI 2022 workshop)**. [core rank A*]
- Mohammad Imrul Jubair, Md Masud Rana, Md Amir Hamza, **Mohsena Ashraf**, Fahim Ahsan Khan, Ahnaf Tahseen Prince, *Altering facial expression based on textual emotion*, Ac-

cepted at International Conference on Computer Vision Theory and Applications **(VISAPP 2022)**. [core rank B]

- Shithi Maitra, Md. Rakib Ahamed, Md. Nazrul Islam, MD Abdullah Al Nasim, **Mohsena Ashraf**,*A Soft Computing Based Customer Lifetime Value Classifier for Digital Retail Businesses*, Presented at IEEE UEMCON 2021. ***[received Best Session Presenter Award]***
- Eshtiak Ahmed, Ashraful Islam, **Mohsena Ashraf**, Md Ibrahim Khan, Atiqul Islam Chowdhury, Asif Karim, *An Investigation Into The Level Of Valence Offered By Different Pointing Devices Against Challenging Tasks*, International Conference on Advanced Computer Science and Information Systems (ICACSIS), October 2020. ***[received Best Session Presenter Award]***
- Atiqul Islam Chowdhury, **Mohsena Ashraf**, Ashraful Islam, Eshtiak Ahmed, Md. Saroar Jaman, Mohammad Masudur Rahman, *hActNET: An Improved Neural Network based Method in Recognizing Human Activities*, $4^{th}$ International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT), October 2020.
- Eshtiak Ahmed, Ashraful Islam, **Mohsena Ashraf**, Atiqul Islam Chowdhury, Mohammad Masudur Rahman, *Internet of Things (IoT): Vulnerabilities, Security Concerns and Things to Consider*, IEEE $11^{th}$ International Conference on Computing, Communication and Networking Technologies (ICCCNT), July 2020.
- Farzana Anowar, **Mohsena Ashraf**, Ashraful Islam, Eshtiak Ahmed, Atiqul Islam Chowdhury, *A Review on Diabetes Self-management Applications for Android Smartphones: Perspective of Developing Countries*, IEEE $2^{nd}$ International Congress on Human-Computer Interaction, Optimization and Robotic Applications, June 2020.
- Tonmoy Hossain, Fairuz Shadmani Shishir, **Mohsena Ashraf**, Md Abdullah Al Nasim, Faisal Muhammad Shah, *Brain Tumor Detection Using Convolutional Neural Network*, International Conference on Advances in Science, Engineering and Robotics Technology, May 2019.
- Faisal Muhammad Shah, Tonmoy Hossain, **Mohsena Ashraf**, Fairuz Shadmani Shishir, MD Abdullah Al Nasim, Md. Hasanul Kabir, *Brain Tumor Segmentation Techniques on Medical Images-A Review*, International Journal of Scientific & Engineering Research, Vol. 10, Issue 2, February 2019.

# EDITORIAL SERVICE

### Reviewer:

- International Journal of Human-Computer Interaction (Taylor & Francis) *[May 2025]*
- The ACM (Association of Computing Machinery) CHI conference on Human Factors in Computing Systems *[October 2024]*
- WIREs Data Mining and Knowledge Discovery Journal *[July 2024]*
- 22nd annual ACM Interaction Design and Children (IDC) Conference (Works-in-Progress Category) *[April 2023]*

- Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization *[September 2022]*

## FELLOWSHIPS

- **Futures Family Fellowship** *[April 2024]* (2500 USD): Received this fellowship at the University of Colorado Boulder, awarded for academic and research excellence, recognizing and supporting student parents balancing education and caregiving responsibilities
- **Endowed Founder's Fellowship** *[October 2023]* (500 USD): Received the Computer Science Endowed Founder's Fund Fellowship at the University of Colorado Boulder, awarded for excellence in academics, research, teaching, and service within the Department
- **Early Career Professional Development Fellowship** *[September 2022]* (1000 USD): From University of Colorado Boulder
- **Departmental Fellowship** *[August 2022]* (4000 USD): In PhD 1st year from University of Colorado Boulder

## AWARDS

- **Outstanding TA Recognition Award** *[May 2023]* (500 USD): Received the Outstanding TA Award from the Computer Science Department, University of Colorado Boulder, in my 1st PhD year, recognizing exceptional TA performance. Responsibilities included testing assignments, conducting study hours, and grading for a graduate-level Computer Vision course. The award was based on evaluations from students, the instructor, and the department.
- Nominated for the **Prime Minister Gold Medal Award 2019** *[December 2021]*: Nominated for the Prime Minister Gold Medal Award by the University Grants Commission, Bangladesh, for achieving the highest CGPA in the faculty at Ahsanullah University of Science and Technology.
- **Course Award** *[March 2020]* (100 Euro): Awarded for securing $4th$ position in the *Practical Cryptography* course under *Facilitating Distance Learning Using Digital Conferencing Facility* project from Asi@Connect organized by *BdREN*
- **Dean's List of Honor** *[July 2019]*: Awarded to the students for achieving a CGPA greater than 3.75 in Bachelor degree at Ahsanullah University of Science and Technology
- **Undergraduate Scholarships in** $4^{th}$ **year** $2^{nd}$ **semester** *[2019]* (17850 BDT): For being $1^{st}$ in the merit position
- **Undergraduate Scholarships in** $4^{th}$ **year** $1^{st}$ **semester** *[2018]* (17850 BDT): For being $1^{st}$ in the merit position

## OTHER RECOGNITIONS

- **Grant for VISAPP 2022** *[December 2021]* (80 Euro): Registration fee
- $4^{th}$ **position among** 152 **international students** *[March 2020]*: in *Practical Cryptography* course organized by *BdREN*
- $1^{st}$ **position among** 152 **students** *[July 2019]*: in *B.Sc.*
- $6^{th}$ **position in Dhaka Division** *[2012]*: at Bangladesh Olympiad on Astronomy and Astrophysics (the national chapter of International Olympiad on Astronomy and Astrophysics)
- $2^{nd}$ **position in Dhaka Division** *[2010]*: at Bangladesh Olympiad on Astronomy and Astrophysics (the national chapter of International Olympiad on Astronomy and Astrophysics)

## MENTORSHIP EXPERIENCE

- **TA Mentor** *[Fall 2024, Fall 2025]*, To help the new TAs of the department with teaching skills and experiences
- **Graduate Peer Mentor** *[Fall 2024, Fall 2023, Fall 2025, Fall 2026]*, To provide personal and professional support, encouragement, and information for new graduate students
- **CU Boulder Prospective Applicant Mentorship Match** *[Fall 2023]*, Assisted in simplifying the application process and evaluating materials submitted by potential PhD applicants

## GUEST LECTURES

- **Counting AI by Hand** *[June 2026]*: Conducted a mini-course containing 3 sessions on Transformer OPs, Agent Tokens, Loop Engineering hosted by *AI by Hand*
- **Advanced AI Agent by Hand** *[September-October 2025]*: Conducted a 7-week mini-course on Advanced AI Agents, covering topics such as *Model Context Protocol (MCP), Agent-to-Agent (A2A) Protocol, Agent Payments (AP2) Protocol, Agent Evaluation, Agentic Knowledge Graph Construction, and Small Language Models for Agents*, hosted jointly by *Packt Publishing* and *AI by Hand*
- **MCP vs Function Calling** *[July 2025]*: Hosted by *AI by Hand*
- **State of the Art Algorithms in LLM** *[June 2025]*: Hosted by *AI by Hand*

## INTERVIEWS & INVITED TALKS

- **Architecture / Building Blocks / Anatomy of LLMs** *[November 2025]*: At *Packt's* Annual Generative AI Conference (GenAI Nexus 2025)

- **Featured in CSGSA Neural Network** *[November 2023]*: In the Interview Series by the Computer Science Graduate Student Association, University of Colorado Boulder. Available here.

## PARTICIPATIONS & OTHERS

- **CSEN PhD Open House General Volunteer** *[February 2024]*, Department of Computer Science, University of Colorado Boulder
- **Participant at Leading with Impact: Artificial Intelligence** *[September 27, 2023]*, (Arranged by CU Boulder)
- **Poster Presenter, Pervasive Personalized Intelligence (PPI) Center** *[April 12, 2023]*, Presented a poster on *'Impact of ChatGPT on CS Graduate Education: A Case Study'*, Spring 2023 Industry Advisory Board Meeting of PPI Center
- **CSEN PhD Open House Core Volunteer** *[February 23 - February 25, 2023]*, Department of Computer Science, University of Colorado Boulder
- **Session Chair, BAIVC Student Symposium (BAIVC23)** *[February 24, 2023]*, University of Colorado Boulder
- **Presenter, BAIVC Student Symposium (BAIVC23)** *[February 24, 2023]*, Presented a Talk on *'A Design Approach for Inclusive fNIRS Study: in Perspective of Muslim Women'*, University of Colorado Boulder
- **Poster Presenter, Annual Research Expo** *[February 23, 2023]*, Presented a poster on *'Diy Graphics Tab: A Cost-Effective Alternative to Graphics Tablet for Educators'*, Department of Computer Science, University of Colorado Boulder
- **Member, Outcome-based Education (OBE) Committee** *[July 2021 – July 2022]*, Department of CSE, Ahsanullah University of Science and Technology
- **Member, Accreditation Committee** *[January 2021 – July 2022]*, Department of CSE, Ahsanullah University of Science and Technology
- **Academic Training, Certificate in eLearning and Teaching CeLT (5th Batch)** *[March 2022]*, Organized by Foundation for Learning, Teaching and Research (fLTR)
- **Member, IT Support Subcommittee** *[December 2020]*, International Conference on Computer and Information Technology, **(ICCIT2020)**
- **Member, Programming Subcommittee** *[2019 - 2022]*, Department of CSE, Ahsanullah University of Science and Technology
- **Participant, Software Category** *[2017]*, *ADEX Engenius*—An inter university tech competition
- **Participant** [2016], *Intra AUST Programming Contest*

## MISCELLANEOUS

- **Youtube Channel** *[**teaching**]*: youtube.com/c/MohsenaAshraf
- **Youtube Channel** *[**cooking**]*: youtube.com/c/RiasCookbook

## REFERENCE


1. **Dr. Tom Yeh**
   Associate Professor
   Department of Computer Science
   University of Colorado Boulder.
   E-mail: `tom.yeh@colorado.edu`